

COMPUTER VISION

SURPRISE TEST

1. write the filtered image using 3X3 mask

$$\begin{bmatrix} 18 & 22 & 33 & 25 & 32 & 24 \\ 34 & 128 & 24 & 172 & 26 & 33 \\ 22 & 19 & 32 & 31 & 28 & 26 \end{bmatrix}$$

0	$\frac{1}{6}$	0
$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}$
0	$\frac{1}{6}$	0

3×3 mask

3 Marks

2. If the size of the averaging filter used to smooth the original image to first image is 9, then what would be the size of the averaging filter used in smoothing the same original picture to



1 Marks

- a). 3
- b). 5
- c). 9
- e). 15

3. The following matrix represents the pixel values of an 8-bit image (r), apply negative transform and find the resulting image pixel values **2 Marks**

Image (r)

100	110	90	95
98	140	145	135
89	90	88	85
102	105	99	115

|

4. A grayscale image $f(x,y)$ is enhanced by applying a filter that directly modifies the pixel values at each coordinate (x,y) . The output image is denoted as $g(x,y)$.

Which of the following expressions correctly represents this spatial domain image processing operation

1Marks

- a) $g(x,y)=T[f(x,y)]$
- b) $f(x+y)=T[g(x+y)]$
- c) $g(xy)=T[f(xy)]$
- d) $g(x-y)=T[f(x-y)]$

5. _____ is the disadvantage of using smoothing filter?

1 Marks

- a) Blur edges
- b) Blur inner pixels
- c) Remove sharp transitions
- d) Sharp edges

6. Consider a 3×3 spatial mask that averages the four closest neighbors of a point (x, y) , but excludes the point itself from the average. **2Marks**

Write an expression for the filter, $h(x, y)$, in the spatial domain

0	$\frac{1}{4}$	0
$\frac{1}{4}$	0	$\frac{1}{4}$
0	$\frac{1}{4}$	0

Assume that the neighborhood of (x, y) can be represented as:

$$\begin{bmatrix} (x-1, y-1) & (x, y-1) & (x+1, y-1) \\ (x-1, y) & (x, y) & (x+1, y) \\ (x-1, y+1) & (x, y+1) & (x+1, y+1) \end{bmatrix}$$